

SUMMARY

PROJECT TITLE: Decoding Customer Value:

A SQL-Driven Retention Strategy

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Tools Used: Python | SQL | Power BI Submission

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This project focused on analyzing customer purchasing behavior, loyalty patterns, promotional dependency, and revenue drivers using Python, SQL, and Power BI to support data-driven business decision-making.

The dataset was cleaned and transformed using Python, where additional engineered features such as customer segments, promotional dependency indicators, and loyalty-related behavioral metrics were created to improve analytical depth. SQL-based customer analysis was then performed to identify revenue trends, customer retention patterns, product category performance, and regional purchasing behavior. Finally, an interactive Power BI dashboard was developed to visualize customer value distribution, promotional dependency, geographic opportunities, seasonal revenue trends, and category-level insights.

The analysis identified three primary customer segments: High Value Organic, Regular, and Discount-Driven customers. Among these, High Value Organic customers demonstrated stronger loyalty behavior, higher average spending, lower promotional dependency, and more sustainable long-term purchasing patterns. In contrast, Discount-Driven customers relied heavily on promotions and exhibited weaker organic retention behavior.

Category-level analysis revealed that products such as Jewelry, Coats, Scarves, and Blouses were associated with stronger repeat purchase behavior, while categories such as Hats, Sneakers, Coat and Sweater demonstrated comparatively higher promotional dependency. Revenue analysis also indicated that Fall generated the strongest seasonal revenue performance, while Summer showed slightly lower purchasing activity.

Additionally, customers selecting premium shipping methods such as Express and 2-Day Shipping displayed relatively higher average spending behavior.

Geographic analysis revealed broad national customer distribution with several regions showing stronger organic demand and lower discount dependency, indicating opportunities for region-specific retention and acquisition strategies.

Based on these findings, the project recommends transitioning from promotion-led growth toward retention-led growth by reducing excessive blanket discounting, strengthening loyalty-based engagement, prioritizing high-value customer segments, and implementing more personalized retention strategies. A phased promotional sunset plan combined with customer-centric retention initiatives can help improve long-term profitability, customer lifetime value, and overall business sustainability.